THE TWO GEOMETRIC FIBRE-COUNT FUNCTIONS OF A LOCALLY QUASI-FINITE MAP AGREE AND ARE BOUNDED ON QUASI-COMPACT SOURCES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 21959-21969 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 21971-21978, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Glue lemma for the weighting machinery of the chapter's integral-of-weightings section (next card's max-int-w builds on this boundedness).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas base-change-fibres-nr-geometrically-irreducible-components and -connected-components (the two definition/compatibility results for n_{X/Y}); Morphisms, lemmas locally-quasi-finite-fibres, characterize-universally-bounded, locally-quasi-finite-qc-source-universally-bounded.

Source reference. Original lemma 055F: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7905–7918. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0556: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7453–7466. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03J5: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21978–21978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06RT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21978–21978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03JA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 21978–21978. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism. Then*

(1) *the functions $n_{X/Y}$ of Lemmas 0556 and 055F agree,*
(2) *if X is quasi-compact, then $n_{X/Y}$ attains a maximum $d < \infty$.*

Proof. Agreement of the functions is immediate from the fact that the (geometric) fibres of a locally quasi-finite morphism are discrete, see Morphisms, Lemma 06RT. Boundedness follows from Morphisms, Lemmas 03J5 and 03JA. □